

Special instructions

- This exam consists of **three parts**. The mark breakdown is as follows:

Part	A	B	C			
Question	1	2	3	4	5	Total
Marks	15	15	10	10	10	60

- **Unless a question indicates a numeric value is required**, numeric answers may be left as numeric expressions if desired (e.g., if the correct answer was 2, an answer of ‘ $1 + 1$ ’ would receive the same credit as an answer of ‘2’). If you express answers as numeric values these should be accurate to at least three decimal places.
- If you are asked to identify the type of a variate, choose from Continuous, Discrete, Categorical, Ordinal, or Complex.
- If you believe insufficient information is provided in a question to determine the requested answer, you may write ‘not enough information’, ‘NEI’, or similar. If the question indicates you should justify your answer, this justification is still required even if your answer is ‘not enough information’.
- Additional blank pages are provided at the end of the exam for rough work and/or additional space for answers. If you wish any work on these pages to be graded, you must clearly indicate this on the corresponding question page.
- Good luck :)

Part A [15 marks]

Question 1 [15 marks]

(a) Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) Which of these numerical summaries is a measure of location?

- A Sample median
- B Interquartile range
- C Sample skewness
- D Sample standard deviation

(ii) What is the range of an empirical cumulative distribution function?

- A $[0, 1]$
- B $(-\infty, +\infty)$
- C It depends on the sample size
- D It depends on the variate being measured

(iii) Which of these probability distributions is discrete?

- A Exponential
- B Gaussian
- C Poisson
- D Normal

(iv) Which of these is NOT included in the five number summary of a sample?

- A The sample mean
- B The sample 50th percentile
- C The minimum
- D The upper quartile

(v) If $L(\theta)$ is the likelihood function and $\hat{\theta}$ the maximum likelihood estimate, then what does $L(\hat{\theta})$ equal?

- A 0
- B 1
- C θ
- D More information is required

(b) [5 marks] Questions (i)-(v) are each worth one mark with no part marks available. You are not required to show any working for calculations.

A sample of size $n = 5$ contains the values $-4, -2, 1, 2, 4$. These data are stored in R in an object named `y`. You are given the following R code and output:

```
> y <- c(-4, -2, 1, 2, 4)
> var(y)
[1] 10.2
```

(i) What is the sample mean of `y`? 0.2

(ii) What is the sample standard deviation of `y`? $\sqrt{10.2}$

(iii) Is the sample skewness of `y` positive, negative or zero? Negative

Note: The sign of the sample skewness is the same as the sign of $\sum_{i=1}^n (y_i - \bar{y})^3$. For these data, that is $-4.2^3 + -2.2^3 + 0.8^3 + 1.8^3 + 3.8^3$. This could be calculated explicitly, or inferred by the properties of cubic powers (e.g., 4.2^3 is much larger than 3.8^3 , 2.2^3 is larger than 1.8^3 , and 0.8^3 is relatively small. Alternatively, we could note that the sample mean is less than the sample median, and while this does not guarantee negative skewness, it's another clue to the answer.

(iv) What number would be output if you ran the command `quantile(y, prob = 0.5)`?

1

(v) Give one reason why it is unlikely these data were sampled from a Gaussian distribution.

They are integers.

(c) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. You are not required to show any working for your calculations. No part marks are available.

Suppose $y_1, y_2, \dots, y_{25}$ is an observed random sample from a probability distribution with corresponding independent and identically distributed random variables $Y_1, Y_2, \dots, Y_{25}$. You are told that the minimum value in the sample is 1, the maximum value in the sample is 10, and that:

$$\sum_{i=1}^{25} y_i = 100 \quad \sum_{i=1}^{25} (y_i - \bar{y})^2 = 600$$

where $\bar{y}$ denotes the sample mean.

(i) Suppose that $Y_1, Y_2, \dots, Y_{25} \sim G(\mu, \sigma)$. What is the maximum likelihood estimate of μ ?

$$\bar{y} = \frac{100}{25} = 4$$

(ii) Suppose $y_1, y_2, \dots, y_{25}$ is an observed random sample from a $G(\mu, \sigma)$ distribution. What is the maximum likelihood estimate of σ ?

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2} = \sqrt{\frac{600}{25}} = \sqrt{24}$$

(iii) Suppose that $Y_1, Y_2, \dots, Y_{25} \sim \text{Exponential}(\theta)$. What is the maximum likelihood estimate of θ ?

$$\bar{y} = 4$$

(iv) Suppose that $Y_1, Y_2, \dots, Y_{25} \sim \text{Exponential}(\theta)$. What is the maximum likelihood estimate of the variance of Y_4 ?

16. Note: The variance of an Exponential random variable is equal to the square of the expectation. The invariance property gives us that this estimate is therefore $4^2 = 16$. (However, an answer of 16 is sufficient for the mark.)

(v) Suppose that $Y_1, Y_2, \dots, Y_{25} \sim \text{Uniform}(0, \theta)$ (where this denotes the continuous Uniform distribution). What is the maximum likelihood estimate of θ ?

10

Note: See Chapter 2, Problem 24 in the Course Notes.

Part B [15 marks]

Part B begins on the following page. All questions in Part B are based on Assignment 1. All variate names refer to those in the Assignment Dataset Information file. All R commands assume the assignment dataset has been stored in R in an object named `mydata`. You may further assume that the variate `subject.age.log` has been created (as a variate within `mydata`) as described in Analysis 2 of Assignment 1.

Important: Unless otherwise stated, you should assume that a question is asking about the entire sample, and not one specific city.

Question 2 [15 marks]

(a) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

Note: Some explanations/comments are provided on the following page.

(i) What type of variate is `day.of.week`?

- A Continuous
- B Discrete
- C Categorical
- D Complex

(ii) Which of these R commands would generate a histogram of the `subject.age` variate in San Antonio in the assignment dataset?

- A `hist(mydata$subject.age[mydata$city == "no"])`
- B `plot(mydata$subject.age[mydata$city != "no"])`
- C `barplot(mydata$subject.age)`
- D None of the above commands would return the desired output

(iii) Which of these R commands would calculate the sample correlation between `subject.age` and `vehicle.year` in the assignment dataset?

- A `cor(mydata$vehicle.year, mydata$subject.age)`
- B `cor(mydata$subject.age, mydata$vehicle.year)`
- C Both A and B could be used.
- D Neither A nor B could be used.

(iv) What is the output of running the command

```
dim(mydata)
```

- A [1] 950
- B [1] 950 16
- C [1] 16 950
- D [1] 16

(v) What is the output of running the command

```
sum(mydata$vehicle.colour == "black")
```

- A [1] 0
- B [1] 193
- C [1] 950
- D [1] 1193

Notes on the preceding questions:

(i) There is an implied ordering but it is not ordinal because days ‘wrap around’: we could order the days Sunday to Saturday, Monday to Sunday, etc. Consider an ordinal variate with values Disagree, Neutral, Agree; we could not order this as (e.g.) Neutral, Agree, Disagree.

(ii) Option A only includes New Orleans. Option B does not generate a histogram.

(iii) Sample correlation is unaffected by variate order (if you swap the axes on a scatter plot, the sample correlation would remain the same).

(iv) The dataset contains 950 observations. The `dim()` command lists the number of rows then columns. It is 16 columns because we have 15 columns to begin with, and have added the `subject.age.log` variate.

(v) This can be deduced from knowledge of the sample size.

(b) [5 marks] Analysis 1c in Assignment 1 was the following:

- 1c. For each city (San Antonio and New Orleans) generate a barplot corresponding to the number (not percentage) of total stops within each category of your chosen variate (`subject.race` or `vehicle.make`). As with the table, the 'other' category should be placed last.

Suppose that two students attempted this question as follows:

Student A chose to analyze the `subject.race` variate and used the following R command (and only the following R command) to answer Analysis 1c:

```
> barplot(table(mydata$subject.race), ylim = c(0, 50), xlab = "Frequency")
```

Student B chose to analyze the `vehicle.make` variate and used the following R command (and only the following R command) to answer Analysis 1c:

```
> barplot(table(mydata$vehicle.make), ylim = c(0, 50), xlab = "Frequency")
```

For Student A *or* Student B, identify five mistakes they have made in their R code. For each mistake, you should briefly describe what problem would arise in the resulting output. You do *not* need to explain how the code could or should be corrected, but you may do so if you feel it aids in your explanation. Note that simply providing a corrected R command is not sufficient to receive full credit: your answer must describe why the preceding code does not produce the desired output.

1 mark for each of the following (with a maximum of 5), which are the same for either command.

- There is no title.
- The plot would be for the entire sample, not each city.
- The x -axis label is incorrect.
- The y -axis label is missing.
- The y -axis range is too small for the sample.
- The 'other' category will not be last.

(c) [5 marks] Consider each of the following statements based on the assignment. For each statement, discuss whether you agree or disagree, providing a brief justification for your answer in 1-2 sentences.

(i) “Based on the scatterplot of `subject.age` and `vehicle.year` we can conclude that the range of `subject.age` is larger than the range of `vehicle.year`.”

Agree. We can see that the range of `subject.age` is over 70 years, while the range of `vehicle.year` is less than 40 years. [1 mark answer.]

(ii) “The relative frequency histogram plots prove that the Gaussian model is more appropriate for `subject.age.log` than `subject.age`.”

Disagree. While the Gaussian distribution appears to fit the histogram for `subject.age.log` more closely than for `subject.age` this is one of many comparisons we can make. Moreover, we can never ‘prove’ the greater suitability of a model in such a context. [2 mark answer.]

(iii) “15.3% of people in San Antonio drive a Ford.”

Disagree. While it is true that 15.3% of traffic stops in the dataset were of people driving a Ford, the statement here is more broad/ambiguous. As such, we cannot agree or disagree with this statement without further context/explanation. [2 mark answer.]

Part C [30 marks]

Question 3 [10 marks]

Every morning for three weeks, a statistics professor sets a five minute timer while they make a cup of tea. During these five minutes, they stare out of their kitchen window until either (i) they see a squirrel; or (ii) the five minute timer elapses. If they see a squirrel, they write down how many seconds they waited before seeing the squirrel, rounded to the nearest second, and then stop looking for squirrels. If they do not see a squirrel within five minutes, they write down a time of 301 seconds for that day.

The professor records 21 ‘squirrel times’, storing the data in R in a table named `mydata` with three columns: `day` (the day number, with 1 indicating the first day of the study and 21 the last day of the study), `day.of.week` (the day of the week), and `squirrel` (either the time in seconds until the first squirrel was seen, or 301 seconds if no squirrel was seen). You are given the following R commands and output:

```
> head(mydata)
  day day.of.week squirrel
1   1      Monday     142
2   2     Tuesday      25
3   3   Wednesday       1
4   4    Thursday      70
5   5     Friday      14
6   6    Saturday       9
> summary(mydata$squirrel)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    1      24      70     96    145    299
> var(mydata$squirrel)
[1] 8216.9
```

(a) [1 mark] What type of variate is `squirrel`?

Continuous.

Note: An answer of ‘continuous or discrete’ with some justification (e.g., noting that we can argue it is ambiguous depending on modelling choices) can also receive this mark.

(b) [1 mark] What is the interquartile range for `squirrel` in the professor’s sample?

$145 - 24 = 121$ or $[24-145]$ (or similar).

(c) [2 marks] On how many days in the sample did the professor not see a squirrel during the five minute timer? Briefly justify your answer.

0 [1 mark for 0] Because the maximum is 299, and not 301, we know the professor saw at least one squirrel each day. [1 mark for justification]

Let $s_1, s_2, \dots, s_{21}$ denote the values of `squirrel` in the professor's sample. Assume these are realizations of a random sample $S_1, S_2, \dots, S_{21}$ where $S_i \sim \text{Exponential}(\theta)$, $i = 1, 2, \dots, 21$, independently. You are reminded that $E[S_i] = \theta$.

(d) [1 mark] Give a numeric value or expression for $\sum_{i=1}^{21} s_i$.

$96 \times 21 = 2016$

(e) [2 marks] Based on the information given in the question, give one reason why the proposed model is not appropriate. Your answer should include a brief justification.

Various possible answers, including:

- All times in excess of 300 seconds are recorded as 301 seconds [2 marks]
- The observations are not independent because (e.g.) they occurred on consecutive days. [2 marks, 1 mark for mentioning independence, 1 mark for justification.]

(f) [3 marks] What is the maximum likelihood estimate of the standard deviation of S_i ? Carefully justify your answer, but you may use any results from the course without proof.

The maximum likelihood estimate of θ is 96.

By the invariance property, the maximum likelihood estimate of the standard deviation of S is also 96.

1 mark for stating/indicating the m.l.e. of θ is 96 Note: answer must explicitly note that this is the m.l.e. in some way (e.g. $\hat{\theta} = 96$).

1 mark for stating 'invariance property'.

1 mark for correctly linking answer to standard deviation of the Exponential distribution.

Question 4 [10 marks]

Alex and Blake are two students working on co-op for *Brot Summer*, an expensive German sandwich shop. As they approach the annual celebration of Oktoberfest, the company emails a voluntary survey to its customers. Upon completion, the customer is asked to share the survey with their family and friends. 50 surveys are completed, with data recorded on each customer's age, number of visits, spend on their last visit, and whether they own a dog. Alex prepared a summary of the survey data as follows, where SD = standard deviation:

Variate	Description	Sample Statistics
Age	The customer's age, in years.	Range: 19-81 Mean: 45.2 Median: 38.9 SD: 10.7
Visits	The number of visits the customer made to the shop in the previous month.	Range: 0-8 Mean: 3.7 Median: 4 SD: 2.0
Spend	How much money the customer spent the last time they visited the shop, in Canadian dollars.	Range: \$0.99-\$234.17 Mean: \$17.25 Median: \$9.50 SD: \$52.50
Dog	Whether the customer has a pet dog ('yes' or 'no').	Yes: 14 No: 36

(a) [2 marks] Do you think it is reasonable to assume these data represent independent observations? Briefly justify your answer.

No. Surveys may not be independent (for example, friends and family may not have independent behaviours with regards to visiting the shop, or how much they spend).

Let D = the number of customers who own a dog in a random sample of n customers. Assume $D \sim \text{Binomial}(n, \delta)$, $0 < \delta < 1$, where d denotes the observed value of D . Write $L(\delta) = P(D = d; \delta)$ for the likelihood function of δ based on the above sample. Write $R(\delta)$ for the corresponding relative likelihood function. Write $\hat{\delta}$ for the maximum likelihood estimate of δ based on this sample.

(b) [2 marks] For each of the following statements, clearly indicate whether it is true, false, or if there is not enough information to determine whether it is true or false (NEI).

(i) $\hat{\delta} = 0.14$: True False NEI It is $14/50 = 0.28$

(ii) $R(\hat{\delta}) > L(\hat{\delta})$: True False NEI

$R(\hat{\delta}) = 1$, $L(\delta)$ is a probability and must be less than 1.

(iii) $P(\delta = \hat{\delta}) = \binom{50}{14} \hat{\delta}^{14} (1 - \hat{\delta})^{36}$: True False NEI

δ is a constant, so $P(\delta = \hat{\delta})$ is either 0 or 1.

(iv) $\delta \neq 1$: True False NEI

(c) [3 marks] For which of the four variates do you think a Poisson model would be most appropriate based on the given information? Justify your answer in 1-2 sentences. Note: ‘Not enough information’, or similar, is not a valid response to this question.

Visits [1 mark] The Poisson model is appropriate for discrete variates, such as ‘Visits’. [1 mark] The other variates are not discrete [1 mark].

Alex’s colleague, Blake, sends the following email and plot:

Hi Alex!

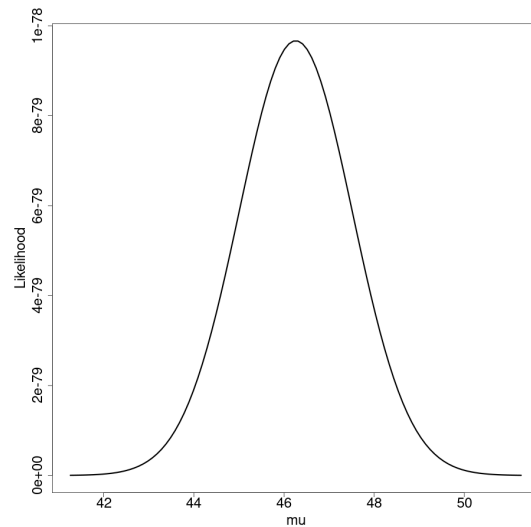
I thought a Gaussian model might be appropriate for the Age variate.

I did some math and came up with the likelihood function for μ assuming a $G(\mu, \sigma)$ model for these data.

This plot looks a lot like a Gaussian distribution to me, so that means the Gaussian model is a good fit to the data, do you agree?

Let me know!

-B



(d) [3 marks] Do you agree with Blake that this plot suggests a Gaussian model is appropriate? Justify your answer and briefly describe one method Blake could use to assess whether a Gaussian model is appropriate.

I do not agree. A likelihood function plot does not tell us anything about whether a Gaussian model is appropriate; it is based on the assumption a Gaussian model is appropriate. [2 marks] The third mark is for any valid Gaussian check method, including (but not limited to) comparing a histogram to a PDF; comparing an ecdf to a CDF; seeing if 95% of values fall within 2 standard deviations of the mean; seeing if the median and mean are similar. (While Q-Q plots were not included in the Midterm 1 material, it would also be acceptable to mention examining a Q-Q plot here!)

Question 5 [10 marks]

A random variable Y takes on the values 1, 2, or 3 according to a discrete probability function $f(y; \theta)$, which is a function of a single unknown parameter θ .

Suppose the probabilities of each value may be written as:

$$P(Y = y; \theta) = \begin{cases} f_1(\theta) & y = 1 \\ f_2(\theta) & y = 2 \\ f_3(\theta) & y = 3 \end{cases}$$

and 0 otherwise, where $f_1(\theta)$, $f_2(\theta)$, and $f_3(\theta)$ are each distinct functions of θ , and $0 < \theta \leq 0.5$.

$n = 8$ values are drawn independently from this distribution, forming the following sample:

1 2 2 2
3 3 3 3

(a) [2 marks] Write down an expression for the likelihood function for this sample in terms of $f_1(\theta)$, $f_2(\theta)$, and $f_3(\theta)$.

$P(Y_1 = 1, Y_2 = 2, \dots, Y_8 = 8) = f_1(\theta) \times f_2(\theta)^3 \times f_3(\theta)^4$ [2 marks for fully correct answer. 1 mark for $P(Y_1 = 1, Y_2 = 2, \dots, Y_8 = 8)$.]

(b) [3 marks] Suppose you are told that the likelihood function for a second sample is

$$L(\theta) = 3\theta^5 + 5\theta^3 + 1$$

(Note: Do not use this to inform your answer to part (a).) Using this likelihood function, what is the maximum likelihood estimate of θ based on this second sample? Carefully justify your answer. Hint: Consider the range of possible values for θ .

Differentiating: $L'(\theta) = 15\theta^4 + 15\theta^2$. Solving for 0 we have $15\hat{\theta}^4 + 15\hat{\theta}^2 = 0$. This is increasing in θ , and so is maximized at $\hat{\theta} = 0.5$ as this is the maximum value of θ .

1 mark for differentiating correctly. 0.5 marks for solving to 0. 1 mark for stating this is increasing in θ . 0.5 marks for concluding $\hat{\theta} = 0.5$.

Note: Full marks possible without differentiating and solving to 0, if it is observed that the function is increasing in θ (leading to the correct answer).

Part (c) concerns a different probability distribution, unrelated to parts (a) and (b).

Let $y_1, y_2, \dots, y_n$ be independent, identically distributed observations drawn from a continuous probability distribution with parameter θ . You are told that the probability density function for this distribution is:

$$g(y; \theta) = y^\theta \log(y^\theta) \quad 0 < y < 1 \text{ and } \theta > 0$$

Note: This is not a real probability density function, but you may assume it has all the properties necessary in order to solve the following questions, and as such may answer the following under the assumption it is a real probability density function. In particular, you may assume that the likelihood and log-likelihood functions are maximized for the value of θ at which their first derivatives equals 0.

(c) [1 mark] Which of statements A-D is true? Circle your answer, then provide a brief justification:

$$A : \sum_{i=1}^n \log(y_i) < 0 \quad B : \sum_{i=1}^n \log(y_i) \leq 0 \quad C : \sum_{i=1}^n \log(y_i) \geq 0 \quad D : \sum_{i=1}^n \log(y_i) > 0$$

Option A [0.5 marks] Because $0 < y_i < 1$, then $\log(y_i) < 0$ for all possible values, so its sum must also be (strictly) negative. [0.5 marks for justification]

A sample of data $y_1, y_2, \dots, y_n$ is observed.

(d) [4 marks] Find the maximum likelihood estimate of θ , expressed as a function of the observed data. Carefully show all your steps.

Likelihood:

$$\begin{aligned} L(y) &= \prod_{i=1}^n [y_i^\theta \log(y_i^\theta)] &= \prod_{i=1}^n [y_i^\theta \theta \log(y_i)] \\ &= \left(\prod_{i=1}^n y_i \right)^\theta \theta^n \prod_{i=1}^n \log(y_i) \\ &\propto \left(\prod_{i=1}^n y_i \right)^\theta \theta^n \end{aligned}$$

$$\text{Log-likelihood: } \ell(y) = \theta \log \left(\prod_{i=1}^n y_i \right) + n \log \theta = \theta \sum_{i=1}^n \log(y_i) + n \log \theta$$

$$\text{Differentiating w.r.t. } \theta: \ell'(y) = \sum_{i=1}^n \log(y_i) + \frac{n}{\theta}$$

Solving for zero:

$$\begin{aligned} \sum_{i=1}^n \log(y_i) + \frac{n}{\hat{\theta}} = 0 &\implies \hat{\theta} \sum_{i=1}^n \log(y_i) + n = 0 \\ &\implies \hat{\theta} = \frac{-n}{\sum_{i=1}^n \log(y_i)} \end{aligned}$$

1 mark for simplified likelihood expression.

1 mark for taking logs correctly.

1 mark for differentiating correctly.

1 mark for solving to 0 to find the m.l.e.

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